

SARTHAK BISWAS

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Summary

AI Engineer, model fine-tuning (SFT, GRPO, DPO, LoRA), training dynamics debugging, and deep learning systems built from scratch.

Experience

Doubtflix (AIPrep) | FastAPI, PostgreSQL, Docker, S3 | HF, Unsloth AI

Remote

AI Engineer

Nov 2025 – Apr 2026

- **AI video generation pipeline** (0 to 1) using **Manim** + **ElevenLabs TTS**, multilingual narration across **14+ languages**.
- **Multi-stage LLM pipeline** with model routing, tool calling, and async task execution.
- **SFT/DPO data pipelines** using **Unsloth AI**, curated datasets across 13 subjects for domain-specific fine-tuning.
- **RAG-powered search** and production APIs, contextual retrieval across educational content.
- **Personalized recommendation engine** using **collaborative filtering**, vector embeddings, weighted ranking with cold-start handling.

Projects

LLM Stock Trader, Fine-Tuning & RL Alignment | *DeepSeek 7B, Unsloth, LoRA, TRL, GRPO*

GitHub | HF

- **15 fine-tuning experiments** (SFT → GRPO) on DeepSeek 7B with **LoRA r=16** (q/k/v/o), lr=5e-6. Score: **0.300 → 0.537** (+79%).
- Diagnosed 3 training failures: **HOLD collapse** (data imbalance), **reward hacking** (format gaming), **KL catastrophe** (KL=4.2).
- **12K reverse-distilled** SFT dataset, outperformed larger noisy sets. Asymmetric reward decomposition to prevent spec gaming.

Stock Trader RL Environment (OpenEnv) | *Python, FastAPI, Pydantic, Docker, HF Spaces*

GitHub | HF

- **Meta PyTorch OpenEnv Hackathon** finalist, top 800 of 32,000+ teams. RL environment for daily trading on NIFTY stocks.
- Three difficulty tiers, regime gates, FeatureEngine (RSI, MACD, Bollinger), Graders (Sharpe, discipline, risk).
- **1.22M-param causal transformer** world model (MDN, 3 Gaussians) as synthetic training environment for GRPO.

ETA Prediction Engine | *Python, PyTorch, LightGBM, MLflow, Docker*

GitHub | HF

- **3-model ensemble** (neural net + LightGBM + FT-Transformer) on **37M trips**, **253s MAE**, **28% better than XGBoost** baseline, inference under 4ms on CPU.
- **26 features** with **Bayesian-shrunk zone-pair statistics** across **6 traffic-regime time buckets**, fallback hierarchy for unseen pairs, zone-pair stats alone beat XGBoost with zero ML.
- **Dual-branch embedding network** (560K params) with learned zone embeddings, **FT-Transformer from scratch** (406K → 169K params), **ONNX Runtime** inference from 11ms to 4ms.

Autonomous Trader Agent | *Python, FastAPI, PostgreSQL, Docker, Zerodha API*

GitHub | Demo

- Autonomous trading on **NIFTY stocks**, **cross-sectional reversal scoring** with **IC +0.020** (large-cap), **+0.025** (midcap).
- **3-state regime classifier** (Bull/Neutral/Weak) with **adaptive confidence scoring**, **8.6% CAGR**, **60% win rate** over 5.4 years.
- **A/B pipeline testing**, risk controls (**kill switches**, **regime gates**, **drawdown dampening**), state persistence to **Neon** Postgres.

Education

Rajasthan Technical University

B.Tech in Information Technology

Kota, Rajasthan

Technical Skills

Languages: Python, TypeScript

AI / ML: PyTorch, HF Transformers, Scikit-learn, Unsloth AI, vLLM

Techniques: LLM Fine-tuning, Reinforcement Learning, Deep Learning, RAG, Feature Engineering, Data Pipelines

Data / Libraries: Pandas, NumPy, OpenCV, NLTK, Matplotlib

Backend / Databases: FastAPI, Celery, Redis, PostgreSQL, Vector Databases

Cloud / DevOps / MLOps: AWS/GCP, DigitalOcean, Docker, CI/CD, GitHub Actions